

AI/ML COMPETITION

Netflix Recommendation System

Personalized Content Discovery
Using Collaborative Filtering & SVD

3M+

Ratings

374K

Users

564

Movies

0.989

Best RMSE

Problem Statement & Dataset



The Challenge

- Netflix offers 10,000+ titles — users can't browse them all
- Content overload leads to decision fatigue and churn
- Personalized recommendations drive 80% of Netflix viewing
- Goal: Predict ratings & surface relevant titles for each user



Netflix Prize Dataset

3,000,000

Total Ratings

374,340

Unique Users

564

Unique Movies

3.62 / 5.0

Avg. Rating

98.43%

Matrix Sparsity

Exploratory Data Analysis

Rating Distribution



INSERT GRAPH HERE

*Ratings skew positive
(3–5 stars dominate)*

User Activity



INSERT GRAPH HERE

*Power-law distribution
in user review count*

Movie Popularity



INSERT GRAPH HERE

*Few titles account for
majority of ratings*

⚠ Matrix Sparsity:

98.43%

— Only 1.57% of user-movie combinations are rated. This drives the need for matrix factorization.

Recommendation Approaches



Item-Based Collaborative Filtering

Memory-Based Approach

- Computes cosine similarity between item rating vectors
- Finds K most similar movies a user has already rated
- Weighted average of neighbor ratings = predicted score
- No model training — predictions at query time
- Interpretable: 'Users who liked X also liked Y'

RMSE: 1.0723



Singular Value Decomposition

✓ SELECTED

Model-Based Approach

- Decomposes rating matrix $R = U \times \Sigma \times V^T$
- Learns latent user & item factors (embeddings)
- Captures hidden taste dimensions automatically
- Gradient descent + regularization during training
- Scales to millions of users; generalizes to new data

RMSE: 0.9889 ◀ Best Model

Methodology & Training Pipeline

01



**Data Loading
& Cleaning**

3M ratings
parse & deduplicate

02



**Train/Test
Split**

80% train
20% holdout

03



**Build Rating
Matrix**

Users × Movies
sparse format

04



**SVD
Factorization**

k=50 factors
SGD optimizer

05



**Predict &
Evaluate**

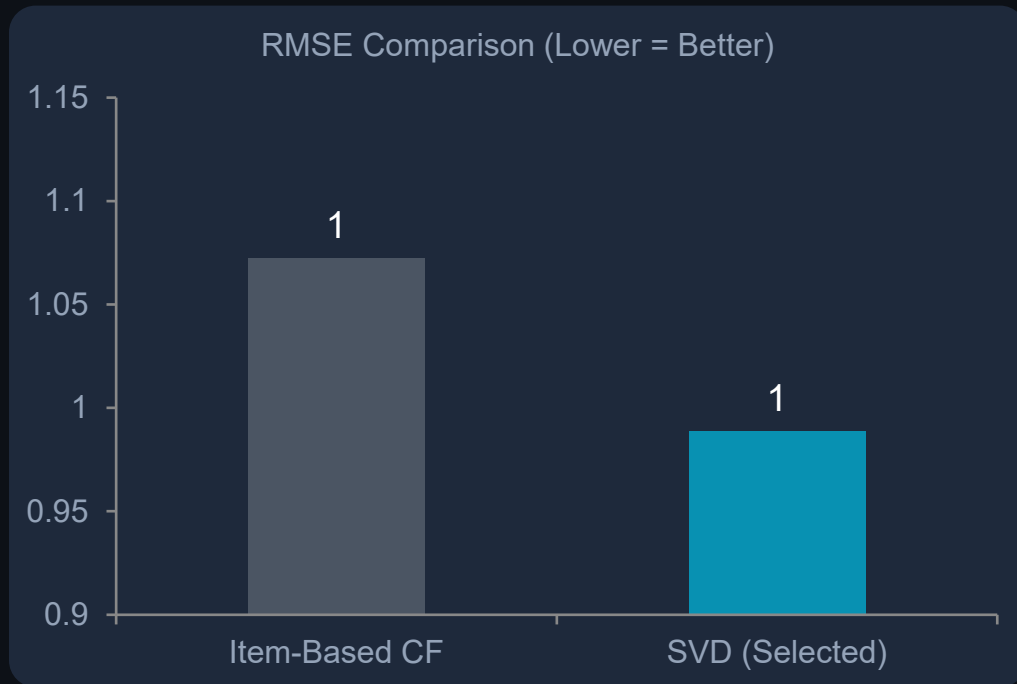
RMSE & MAP@10
metrics

SVD Core Equation:

$$\mathbf{R} \approx \mathbf{U} \times \mathbf{\Sigma} \times \mathbf{V}^T$$

U = User factors | Σ = Singular values | V = Item factors | k = 50 latent dims

Results & Model Comparison



↑ Replace with your actual RMSE comparison graph

1.0723

Item-Based CF — RMSE

0.9889

SVD Model — RMSE ← Best

7.78%

Improvement — RMSE reduction

MAP@10: Evaluated for top-10 ranking quality

Top-10 Recommendation Examples



INSERT SCREENSHOT HERE

Top-10 Recommendations Table
(from your notebook output)

What the Model Learned

Genre Coherence

Recommendations cluster around user's preferred genres

Latent Patterns

Model discovers hidden taste dimensions without labels

Rating Calibration

Predicted scores align with actual rating scale

Diversity

Top-10 includes variety — not just blockbusters

Conclusion & Future Work

Key Findings

- ✓ SVD outperforms Item-Based CF (RMSE 0.989 vs 1.072)
- ✓ Matrix factorization handles 98.4% sparsity effectively
- ✓ Latent factor models generalize beyond observed ratings
- ✓ Top-10 MAP@10 confirms ranking quality, not just accuracy

Future Work

- **Deep Learning**
Neural Collaborative Filtering (NCF) & autoencoders
- **Hybrid Models**
Combine content-based + collaborative signals
- **Real-Time**
Online learning with streaming user feedback
- **Explainability**
Generate natural-language recommendation rationale

SVD-based personalization scales to production — delivering the right content to the right user, every time.